

Title: Teacher Digital Competence and Its Influence on the Use of Educational Technology in Blended Learning

ABSTRACT

This study examined teacher digital competence and its influence on the use of educational technology in blended learning. The study focused on teachers' ability to use digital tools for lesson preparation, content delivery, assessment, communication, and student monitoring. A descriptive quantitative research design was used. Data were gathered from selected teachers through a survey questionnaire. The findings showed that teachers generally had a moderate to high level of digital competence, especially in using presentation tools, online communication platforms, and Learning Management Systems. However, some teachers still experienced challenges in using advanced features such as digital assessment tools, analytics, and interactive learning applications. The study found that teacher digital competence influenced how often and how effectively educational technology was used in blended learning. The study recommends continuous teacher training, technical support, and school-based digital learning policies to improve technology integration.

Keywords: teacher digital competence, educational technology, blended learning, digital literacy, learning management system, teacher training, technology integration

CHAPTER 1 INTRODUCTION

Background of the Study

Educational technology has become an important part of teaching and learning. Many schools now use blended learning, where face-to-face instruction is combined with online learning activities. Teachers use different digital tools such as Learning Management Systems, video conferencing platforms, online quizzes, digital presentations, file-sharing tools, and communication applications.

However, the effective use of educational technology depends on the digital competence of teachers. Digital competence refers to the ability of teachers to use technology properly, confidently, and responsibly in teaching. Some teachers are comfortable using digital tools, while others still need training and support. This difference can affect how educational technology is used in blended learning.

In blended learning, teachers are expected to prepare digital materials, manage online activities, communicate with students, give feedback, and monitor student progress. If teachers have low digital competence, they may struggle to use available tools effectively. This may result in limited online interaction, delayed feedback, and less engaging learning activities.

This study focused on teacher digital competence and how it influences the use of educational technology in blended learning.

Statement of the Problem

This study aimed to determine how teacher digital competence influences the use of educational technology in blended learning.

Specifically, it answered the following questions:

1. What is the level of teacher digital competence in terms of:
 - lesson preparation;
 - content delivery;
 - assessment;
 - communication; and
 - student monitoring?
2. What educational technology tools are commonly used by teachers in blended learning?
3. How often do teachers use educational technology in blended learning?
4. What challenges do teachers experience when using educational technology?
5. What recommendations may be proposed to improve teacher digital competence?

Objectives of the Study

This study aimed to:

1. Determine the level of digital competence among teachers.
2. Identify the educational technology tools commonly used in blended learning.
3. Assess how teacher digital competence influences technology use.
4. Identify challenges experienced by teachers in using digital tools.
5. Recommend ways to improve teacher training and technology integration.

Significance of the Study

This study may benefit the following:

Teachers. The study may help teachers identify areas where they need improvement in using educational technology.

Students. Students may benefit from more organized, interactive, and effective blended learning activities.

School Administrators. The findings may help administrators design teacher training programs and improve digital learning support.

Future Researchers. This study may serve as a reference for future studies about digital competence, blended learning, and educational technology.

Scope and Delimitation

This study focused on selected teachers who were using blended learning in their classes. It examined teacher digital competence, commonly used educational technology tools, frequency of use, and challenges encountered. The study did not directly measure student academic performance.

CHAPTER 2

REVIEW OF RELATED LITERATURE

Teacher Digital Competence

Teacher digital competence refers to the knowledge, skills, and confidence needed to use digital tools in teaching. It includes the ability to prepare digital lessons, present content online, assess students through digital platforms, communicate using technology, and monitor learning progress.

Teachers with strong digital competence are more likely to use technology in meaningful ways. They can create interactive learning materials, provide timely feedback, and manage online learning activities more effectively.

Educational Technology in Blended Learning

Educational technology includes digital tools used to support instruction. These may include Learning Management Systems, online quizzes, video conferencing tools, digital presentations, educational applications, and communication platforms.

In blended learning, educational technology helps connect classroom instruction with online learning. It allows students to access resources, submit requirements, participate in activities, and communicate with teachers even outside the classroom.

Challenges in Technology Integration

Although technology can improve learning, teachers may face challenges such as lack of training, unstable internet connection, limited access to devices, difficulty using advanced tools, and lack of technical support. These challenges can affect how often and how effectively teachers use educational technology.

CHAPTER 3

METHODOLOGY

Research Design

This study used a descriptive quantitative research design. This design was appropriate because the study aimed to describe the level of teacher digital competence and the use of educational technology in blended learning.

Participants of the Study

The participants of the study were selected teachers who used blended learning in their classes. They were chosen because they had experience using digital tools for instruction, assessment, communication, and student monitoring.

Research Instrument

The study used a survey questionnaire as the main research instrument. The questionnaire included items about:

1. Teacher profile;
2. Digital competence;
3. Educational technology tools used;
4. Frequency of technology use;
5. Challenges encountered; and
6. Recommendations for improvement.

Data Gathering Procedure

The researcher requested permission to conduct the study. After approval, the survey questionnaire was distributed to the selected teacher-participants. The participants were informed about the purpose of the study. Their responses were kept confidential and used only for research purposes.

After the questionnaires were collected, the responses were organized, tabulated, and analyzed.

Data Analysis

The data were analyzed using frequency, percentage, mean, and ranking. These were used to describe the level of digital competence, commonly used tools, frequency of technology use, and common challenges experienced by teachers.

CHAPTER 4

PRESENTATION, ANALYSIS, AND INTERPRETATION OF DATA

Level of Teacher Digital Competence

The results showed that teachers had a moderate to high level of digital competence. Teachers were most confident in using basic digital tools such as presentation software, messaging applications, video conferencing platforms, and Learning Management Systems.

Teachers showed stronger competence in lesson preparation and communication. Many teachers were able to prepare digital materials, upload files, send announcements, and communicate with students through online platforms.

However, some teachers had lower confidence in using advanced features such as online analytics, automated grading tools, interactive simulations, and digital assessment reports.

Educational Technology Tools Used

The commonly used educational technology tools included:

1. Learning Management Systems;
2. Google Forms or online quiz tools;
3. PowerPoint or digital presentation tools;
4. Video conferencing platforms;
5. Messaging applications;
6. File-sharing platforms; and
7. Online collaboration tools.

The most frequently used tools were those related to communication, file sharing, and content delivery. Tools for analytics and advanced assessment were used less often.

Influence of Digital Competence on Technology Use

The findings showed that teachers with higher digital competence used educational technology more often and more effectively. They were more likely to create online activities, use digital assessments, provide feedback through online platforms, and monitor student progress.

Teachers with lower digital competence tended to use technology only for basic tasks such as sending files, posting announcements, and communicating with students. This shows that digital competence affects not only the frequency of technology use, but also the quality of technology integration.

Challenges Encountered by Teachers

The common challenges encountered by teachers included:

1. Limited training on digital tools;
2. Unstable internet connection;
3. Lack of time to prepare digital materials;
4. Difficulty using advanced technology features;
5. Limited technical support;
6. Student participation issues; and
7. Lack of confidence in using unfamiliar platforms.

These challenges show that technology integration requires continuous support, not just access to digital tools.

CHAPTER 5

SUMMARY, CONCLUSIONS, AND RECOMMENDATIONS

Summary of Findings

This study examined teacher digital competence and its influence on the use of educational technology in blended learning. The findings showed that teachers generally had a moderate to high level of digital competence. They commonly used Learning Management Systems, online communication tools, digital presentations, online quizzes, and file-sharing platforms.

The study also found that teachers with higher digital competence were more likely to use educational technology effectively. They used digital tools not only for communication, but also for assessment, feedback, and student monitoring.

However, teachers still experienced challenges such as limited training, technical issues, unstable internet connection, and lack of confidence in using advanced tools.

Conclusions

Based on the findings, the study concludes that teacher digital competence plays an important role in the effective use of educational technology in blended learning. Teachers who are more digitally competent are better able to integrate technology into their teaching practices.

The study also concludes that access to digital tools is not enough. Teachers also need proper training, technical support, and time to prepare meaningful digital learning activities.

Recommendations

Based on the findings and conclusions, the following recommendations are proposed:

1. Schools should provide regular training on educational technology tools.
2. Teachers should be given support in using advanced LMS features, online assessments, and digital monitoring tools.
3. Technical support should be available to help teachers address problems during blended learning.
4. School administrators should create clear guidelines for the use of educational technology.
5. Teachers should be encouraged to share best practices in using digital tools.
6. Future researchers may conduct a similar study involving student performance or student engagement.

Final Remarks

This study shows that teacher digital competence is an important factor in successful blended learning. When teachers are confident and skilled in using educational technology, they can create more organized, interactive, and effective learning experiences for students.

REFERENCES

References will be added based on the final sources used in the actual study.